THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 1: **Real Numbers** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 2	$2250 = 2 \times 3^2 \times 5^3$
2	**(a)** 22338	LCM $= \dfrac{306 \times 657}{9}$
3	**(b)** always irrational	If it were rational, dividing by the rational would make the irrational rational
4	**(d)** 5	$3125 = 5^5$; the number of places is the larger of the powers of 2 and 5
5	**(d)** $\sqrt{5} + 3$	Rational + irrational is irrational
6	**(d)** no natural number n	$3 = 3$ lacks the prime factor 5
7	**(a)** 5	$480 = 2^5 \times 3 \times 5$
8	**(a)** 2	$3600 = 2^4 \times 3^2 \times 5^2$
9	**(b)** 23460	LCM $= \dfrac{510 \times 92}{2}$
10	**(b)** always irrational	If it were rational, dividing by the rational would make the irrational rational

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$6 = 2 \times 3$ has no factor 5 (1); so 6^n never ends in 0 (1).	1+1
12	LCM $= \dfrac{306 \times 657}{9}$ (1) $= 22338$ (1).	1+1
13	$7 \times 6 + 7 = 7(6 + 1)$ (1) has factors 7 and 7, other than 1 and itself (1).	1+1
14	$14 = 2 \times 7$ has no factor 5 (1); so 14^n never ends in 0 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Assume $\sqrt{2} = a/b$ in lowest terms, $b \neq 0$ (1); then $2b^2 = a^2$, so $2 \mid a^2$ hence $2 \mid a$ (1); put $a = 2c$: $b^2 = 2c^2$ so $2 \mid b$ - contradicts coprime (1).	1+1+1
16	Assume $3 + 2\sqrt{5} = r$ rational (1); then $\sqrt{5} = \dfrac{r-3}{2}$ is rational (1) - contradiction; so it is irrational (1).	1+1+1
17	$26 = 2 \times 13$, $91 = 7 \times 13$ (1); HCF $= 13$, LCM $= 182$ (1); $13 \times 182 = 2366$ verified (1).	1+1+1
	OR: Assume $\sqrt{3} = a/b$ in lowest terms, $b \neq 0$ (1); then $3b^2 = a^2$, so $3 \mid a^2$ hence $3 \mid a$ (1); put $a = 3c$: $b^2 = 3c^2$ so $3 \mid b$ - contradicts coprime (1).	1+1+1
18	Assume $2 + 3\sqrt{7} = r$ rational (1); then $\sqrt{7} = \dfrac{r-2}{3}$ is rational (1) - contradiction; so it is irrational (1).	1+1+1
	OR: $510 = 2 \times 3 \times 5 \times 17$, $92 = 2^2 \times 23$ (1); HCF $= 2$, LCM $= 23460$ (1); $2 \times 23460 = 46920$ verified (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Assume $\sqrt{17} = a/b$, with a, b coprime integers, $b \neq 0$ (1); squaring, $17b^2 = a^2$, so $17 \mid a^2$ and hence $17 \mid a$ (1); write $a = 17c$: $b^2 = 17c^2$, so $17 \mid b$ (1); then 17 is a common factor of a and b, contradicting coprimality (1); so $\sqrt{17}$ is irrational (1).	1+1+1+1+1
	OR: Maximum weight $=$ HCF$(420, 130) = 10$ kg (3); trays $= \dfrac{420}{10} + \dfrac{130}{10} = 42 + 13 = 55$ (2).	3+2
20	Assume $\sqrt{19} = a/b$, with a, b coprime integers, $b \neq 0$ (1); squaring, $19b^2 = a^2$, so $19 \mid a^2$ and hence $19 \mid a$ (1); write $a = 19c$: $b^2 = 19c^2$, so $19 \mid b$ (1); then 19 is a common factor of a and b, contradicting coprimality (1); so $\sqrt{19}$ is irrational (1).	1+1+1+1+1
	OR: Maximum weight $=$ HCF$(300, 175) = 25$ kg (3); trays $= \dfrac{300}{25} + \dfrac{175}{25} = 12 + 7 = 19$ (2).	3+2

Mistakes to avoid:

- Using HCF × LCM = product for **three** numbers; the result holds only for two numbers.
- Taking the *greatest* power for HCF or the *smallest* power for LCM; HCF uses smallest common powers, LCM uses greatest powers.

- Checking the denominator before reducing the fraction to lowest terms when testing for a terminating decimal.
- Assuming the product (or sum) of two irrational numbers is always irrational: $\sqrt{2} \times \sqrt{2} = 2$.
- In irrationality proofs, forgetting to state that a and b are co-prime at the start, which is what gives the contradiction.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 1: **Real Numbers** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 3	$8 = 2^3$; the number of places is the larger of the powers of 2 and 5
2	**(b)** $\sqrt{7} + 2$	Rational + irrational is irrational
3	**(a)** no natural number n	$7 = 7$ lacks the prime factor 5
4	**(c)** 6	$1344 = 2^6 \times 3 \times 7$
5	**(d)** 2	$1176 = 2^3 \times 3 \times 7^2$
6	**(c)** 3024	LCM $= \dfrac{336 \times 54}{6}$
7	**(c)** always irrational	If it were rational, dividing by the rational would make the irrational rational
8	**(a)** 3	$200 = 2^3 \times 5^2$; the number of places is the larger of the powers of 2 and 5
9	**(d)** $\sqrt{3} + 4$	Rational + irrational is irrational
10	**(b)** no natural number n	$9 = 3^2$ lacks the prime factor 5

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	LCM $= \dfrac{336 \times 54}{6}$ (1) $= 3024$ (1).	1+1
12	$7 \times 11 + 7 = 7(11 + 1)$ (1) has factors 7 and 12, other than 1 and itself (1).	1+1
13	$21 = 3 \times 7$ has no factor 5 (1); so 21^n never ends in 0 (1).	1+1
14	LCM $= \dfrac{510 \times 92}{2}$ (1) $= 23460$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Assume $\sqrt{5} = a/b$ in lowest terms, $b \neq 0$ (1); then $5b^2 = a^2$, so $5 \mid a^2$ hence $5 \mid a$ (1); put $a = 5c$: $b^2 = 5c^2$ so $5 \mid b$ - contradicts coprime (1).	1+1+1
16	Assume $5 + 4\sqrt{3} = r$ rational (1); then $\sqrt{3} = \dfrac{r-5}{4}$ is rational (1) - contradiction; so it is irrational (1).	1+1+1
17	$336 = 2^4 \times 3 \times 7$, $54 = 2 \times 3^3$ (1); HCF $= 6$, LCM $= 3024$ (1); $6 \times 3024 = 18144$ verified (1).	1+1+1
	OR: Assume $\sqrt{7} = a/b$ in lowest terms, $b \neq 0$ (1); then $7b^2 = a^2$, so $7 \mid a^2$ hence $7 \mid a$ (1); put $a = 7c$: $b^2 = 7c^2$ so $7 \mid b$ - contradicts coprime (1).	1+1+1
18	Assume $7 + 1\sqrt{2} = r$ rational (1); then $\sqrt{2} = \dfrac{r-7}{1}$ is rational (1) - contradiction; so it is irrational (1).	1+1+1
	OR: $144 = 2^4 \times 3^2$, $198 = 2 \times 3^2 \times 11$ (1); HCF $= 18$, LCM $= 1584$ (1); $18 \times 1584 = 28512$ verified (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Assume $\sqrt{23} = a/b$, with a, b coprime integers, $b \neq 0$ (1); squaring, $23b^2 = a^2$, so $23 \mid a^2$ and hence $23 \mid a$ (1); write $a = 23c$: $b^2 = 23c^2$, so $23 \mid b$ (1); then 23 is a common factor of a and b, contradicting coprimality (1); so $\sqrt{23}$ is irrational (1).	1+1+1+1+1
	OR: Maximum weight $=$ HCF$(252, 180) = 36$ kg (3); trays $= \dfrac{252}{36} + \dfrac{180}{36} = 7 + 5 = 12$ (2).	3+2
20	Assume $\sqrt{29} = a/b$, with a, b coprime integers, $b \neq 0$ (1); squaring, $29b^2 = a^2$, so $29 \mid a^2$ and hence $29 \mid a$ (1); write $a = 29c$: $b^2 = 29c^2$, so $29 \mid b$ (1); then 29 is a common factor of a and b, contradicting coprimality (1); so $\sqrt{29}$ is irrational (1).	1+1+1+1+1
	OR: Assume $\sqrt{3} = x/y$, coprime (1); $3y^2 = x^2 \Rightarrow 3 \mid x$, $x = 3c$, $3 \mid y$ - contradiction (2); if $5 - 2\sqrt{3} = r$ were rational then $\sqrt{3} = (5 - r)/2$ would be rational - contradiction (2).	1+2+2

Mistakes to avoid:

- Using HCF × LCM = product for **three** numbers; the result holds only for two numbers.

- Taking the *greatest* power for HCF or the *smallest* power for LCM; HCF uses smallest common powers, LCM uses greatest powers.
- Checking the denominator before reducing the fraction to lowest terms when testing for a terminating decimal.
- Assuming the product (or sum) of two irrational numbers is always irrational: $\sqrt{2} \times \sqrt{2} = 2$.
- In irrationality proofs, forgetting to state that a and b are co-prime at the start, which is what gives the contradiction.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 1: **Real Numbers** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 5	$2400 = 2^5 \times 3 \times 5^2$
2	**(c)** 5	$4320 = 2^5 \times 3^3 \times 5$
3	**(d)** 1584	$\text{LCM} = \dfrac{144 \times 198}{18}$
4	**(d)** always irrational	If it were rational, dividing by the rational would make the irrational rational
5	**(b)** 6	$1600 = 2^6 \times 5^2$; the number of places is the larger of the powers of 2 and 5
6	**(d)** $\sqrt{11} + 5$	Rational + irrational is irrational
7	**(b)** no natural number n	$21 = 3 \times 7$ lacks the prime factor 5
8	**(a)** 9	$3584 = 2^9 \times 7$
9	**(b)** 3	$2646 = 2 \times 3^3 \times 7^2$
10	**(c)** 760	$\text{LCM} = \dfrac{95 \times 152}{19}$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$5 \times 9 + 5 = 5(9 + 1)$ (1) has factors 5 and 10, other than 1 and itself (1).	1+1
12	$35 = 5 \times 7$ has no factor 2 (1); so 35^n never ends in 0 (1).	1+1
13	$\text{LCM} = \dfrac{144 \times 198}{18}$ (1) $= 1584$ (1).	1+1
14	$3 \times 8 + 3 = 3(8 + 1)$ (1) has factors 3 and 9, other than 1 and itself (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Assume $\sqrt{11} = a/b$ in lowest terms, $b \neq 0$ (1); then $11b^2 = a^2$, so $11 \mid a^2$ hence $11 \mid a$ (1); put $a = 11c$: $b^2 = 11c^2$ so $11 \mid b$ - contradicts coprime (1).	1+1+1
16	Assume $4 + 5\sqrt{11} = r$ rational (1); then $\sqrt{11} = \dfrac{r-4}{5}$ is rational (1) - contradiction; so it is irrational (1).	1+1+1
17	$306 = 2 \times 3^2 \times 17$, $657 = 3^2 \times 73$ (1); HCF $= 9$, LCM $= 22338$ (1); $9 \times 22338 = 201042$ verified (1).	1+1+1
	OR: Assume $\sqrt{13} = a/b$ in lowest terms, $b \neq 0$ (1); then $13b^2 = a^2$, so $13 \mid a^2$ hence $13 \mid a$ (1); put $a = 13c$: $b^2 = 13c^2$ so $13 \mid b$ - contradicts coprime (1).	1+1+1
18	Assume $6 + 2\sqrt{13} = r$ rational (1); then $\sqrt{13} = \dfrac{r-6}{2}$ is rational (1) - contradiction; so it is irrational (1).	1+1+1
	OR: $95 = 5 \times 19$, $152 = 2^3 \times 19$ (1); HCF $= 19$, LCM $= 760$ (1); $19 \times 760 = 14440$ verified (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Assume $\sqrt{31} = a/b$, with a, b coprime integers, $b \neq 0$ (1); squaring, $31b^2 = a^2$, so $31 \mid a^2$ and hence $31 \mid a$ (1); write $a = 31c$: $b^2 = 31c^2$, so $31 \mid b$ (1); then 31 is a common factor of a and b, contradicting coprimality (1); so $\sqrt{31}$ is irrational (1).	1+1+1+1+1
	OR: Assume $\sqrt{5} = x/y$, coprime (1); $5y^2 = x^2 \Rightarrow 5 \mid x$, $x = 5c$, $5 \mid y$ - contradiction (2); if $3 - 4\sqrt{5} = r$ were rational then $\sqrt{5} = (3 - r)/4$ would be rational - contradiction (2).	1+2+2
20	Assume $\sqrt{37} = a/b$, with a, b coprime integers, $b \neq 0$ (1); squaring, $37b^2 = a^2$, so $37 \mid a^2$ and hence $37 \mid a$ (1); write $a = 37c$: $b^2 = 37c^2$, so $37 \mid b$ (1); then 37 is a common factor of a and b, contradicting coprimality (1); so $\sqrt{37}$ is irrational (1).	1+1+1+1+1
	OR: Assume $\sqrt{7} = x/y$, coprime (1); $7y^2 = x^2 \Rightarrow 7 \mid x$, $x = 7c$, $7 \mid y$ - contradiction (2); if $2 - 3\sqrt{7} = r$ were rational then $\sqrt{7} = (2 - r)/3$ would be rational - contradiction (2).	1+2+2

Mistakes to avoid:

- Using HCF × LCM = product for **three** numbers; the result holds only for two numbers.

- Taking the *greatest* power for HCF or the *smallest* power for LCM; HCF uses smallest common powers, LCM uses greatest powers.
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